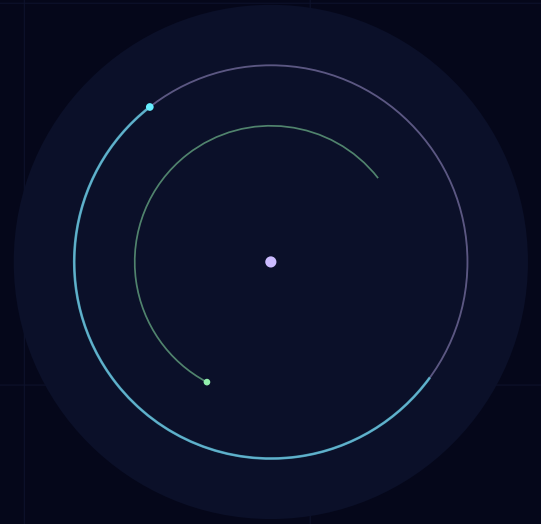




MOONWEAVE AI

RESEARCH · ENGINEERING · GOVERNANCE



FORMAL SEMANTICS AND THEORY-CLOSURE REPORT

Cantilune: A General-Purpose Language and Control Substrate for Agent Orchestration

Cantilune: 通用 Agent 编排系统

A formal-semantics and theory-closure evidence record

PROJECT

Cantilune

VERSION

v0.1

STATUS

PRE-ALPHA

PUBLISHED

2026-07-28

Evidence-bearing document · Formal claims, source, evidence, scope, and review status are versioned.

Document control / 文档控制

Source boundary / 唯一事实源

This report is aligned to the codex/theory - foundation branch of the Cantilune repository. It distinguishes a kernel-checked generic theory from independent review, FCP disposition, product conformance, and any future runtime release. A source revision, evidence revision, or review decision invalidates the affected claims until this record is updated.

Field / 字段	Value / 值
Organization	Moonweave AI
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Proof manifest	formal/proof-obligations.json
Runtime / model	Not applicable — no released runtime or model
Dataset	Not applicable — no training or evaluation dataset
License	Not yet declared for the pre-alpha source tree
Contact	liuzhh268@mail2.sysu.edu.cn
Repository	https://github.com/Moonweave-AI/cantilune/tree/codex/theory-foundation

Abstract / 摘要

ENGLISH

Cantilune is a pre-alpha formal-semantics and orchestration research project. Its proposed substrate represents an orchestration as $\text{CantiluneGraph} = (\mathcal{C}, \mathcal{R})$: a typed symmetric monoidal category for legal composition together with identified string-diagram rewriting events. DAG, Petri, π -calculus, and morphism views are intended as projections of that one object rather than disconnected runtime models. This report records the current theory-closure evidence: a generic core bound to source and evidence commits with status *proved* / *review-pending*. It does not claim a released runtime, SDK, scheduler, product package, benchmark result, or production safety property.

简体中文

Cantilune 是一个 pre-alpha 阶段的形式语义与编排研究项目。其拟议的基础结构将编排表述为 $\text{CantiluneGraph} = (\mathcal{C}, \mathcal{R})$ ：以类型化对称幺半范畴描述合法组合，并以具备身份的字符串图重写事件描述演化。DAG、Petri、 π 演算与态射视图被设计为同一形式对象的投影，而不是彼此脱节的运行时模型。本报告记录当前理论闭包证据：通用核心已绑定到源码与证据提交，状态为 *proved* / *review-pending*。它不宣称已发布运行时、SDK、调度器、产品包、基准结果或生产安全属性。

Keywords / 关键词： agent orchestration, formal semantics, string-diagram rewriting, proof provenance, Lean 4

Executive summary / 执行摘要

This evidence package answers a narrower question than a product release report: what formal object is specified, what theory has a commit-bound proof record, and which claims remain outside that record. The repository’s README, formal specification, proof manifest, and delivery record are the authoritative sources (Moonweave AI, 2026f,g,c,a).

18 / 18

Central obligations

Manifest status: proved

60

Registry operations

Reference P1c operation registry

4

Projection readings

DAG, Petri, pi, morphism

Decision / 决策

No deployment decision. The generic formal core is evidence-bound, but independent review, FCP disposition, ADR acceptance, and product-specific conformance remain pending. Cantilune therefore remains a research and contract-design project, not a runtime release.

- **Formal object:** $\mathcal{C}$ specifies typed sequential and parallel composition; $\mathcal{R}$ specifies identified reconfiguration events.
- **Consistency target:** one source event should have certified counterparts in DAG, Petri, π , and morphism readings, including replay identity and terminal classification.
- **Evidence boundary:** the current record binds the generic core to the verified source and evidence commits; it does not instantiate the planned production packages.
- **Non-claims:** no model integration, tools, I/O, performance benchmark, availability target, or real-world authorization guarantee is established by this theory report.

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Context, claims, and scope

背景、声明与范围

WHY THIS REPORT EXISTS

Cantilune is a research project about an inspectable semantic substrate for orchestration. The repository explicitly describes it as pre-alpha: it is not yet a runtime, SDK, scheduler, or agent product (Moonweave AI, 2026f). This report therefore records a theory and evidence boundary rather than presenting product performance or a release recommendation.

1.1 Problem statement / 问题定义

When several agents, tools, people, services, permissions, sessions, and scarce resources share work, the hard questions are coordination questions: who owns a task, which authority or token is held, when a handoff is valid, and how a change can be replayed. Cantilune proposes that these questions should be visible in one evolving formal object instead of being implicit in prompts, callbacks, and mutable application state (Moonweave AI, 2026f,d).

Insight / 洞察

The unit of accountability is not an LLM response. It is an identified reconfiguration event with its rule, match, derivation data, projection counterparts, and provenance. Whether a future runtime can produce such events is a separate product-conformance question.

1.2 Scope and non-goals / 范围与非目标

Table / 表 1.1: Scope fixed by the current branch. “Planned” is deliberately not reported as shipped.

Dimension	In scope / 范围内	Out of scope / 范围外
Formal core	Typed composition, string-diagram rewriting, event identity, and projection-certificate interfaces	A deployed scheduler or service implementation
Mechanization	Lean 4 source, proof manifest, source integrity, and commit-bound build evidence	Independent mathematical review or governance approval
Reference witnesses	Fixed-signature and normative-family reference executions used to show interfaces are nonempty	Product-specific rules, policies, schemas, or runtime adapters
Interoperability	Intended boundaries for agents, tools, people, MCP/A2A services, and external systems	A claim of live protocol compatibility or provider feature parity
Evaluation	Formal proof status and artifact provenance	Latency, cost, benchmark, reliability, or safety-performance claims

1.3 Claim taxonomy / 声明分类

Every statement in this report belongs to one of the following classes. The classification avoids turning a mechanized interface into a claim about a future product.

Table / 表 1.2: Evidence classes used throughout this report.

Class	Example	Minimum evidence
Specified	$\text{CantiluneGraph} = (\mathcal{C}, \mathcal{R})$ is the proposed formal object	Normative specification and linked RFC
Mechanized	A named theorem has a Lean declaration	Source path, declaration name, and manifest entry
Evidence-bound	A theorem is reported <i>proved</i>	Verified source commit, build evidence, and integrity record
Review-pending	The core is ready for independent review	Explicit absence of independent reviewer, FCP, or ADR decision
Deferred	A production package must supply its own conformance certificate	Product rule family, policy, and reviewed evidence

1.4 Success criterion for this report / 本报告的成功条件

The report is successful only if a reader can separate three statements: what is formally specified, what has a commit-bound kernel-checked record, and what remains to be independently reviewed or instantiated. It does not use aggregate task success, service-level objectives, or deployment readiness as substitutes for those distinctions.

Formal object and system overview

形式对象与系统概览

2.1 The unified object / 统一形式对象

The proposed semantic target is

$$\text{CantiluneGraph} = (\mathcal{C}, \mathcal{R}), \quad (2.1)$$

where $\mathcal{C}$ is a typed symmetric monoidal category describing legal sequential composition ($\circ$), parallel composition ($\otimes$), and symmetry; $\mathcal{R}$ is a set of string-diagram rewriting rules describing identified changes to the collaboration structure (Moonweave AI, 2026g,d). A rule name alone is insufficient to identify an execution: a concrete event includes the selected rule, match/redex, and derivation or fresh-allocation data needed for replay.

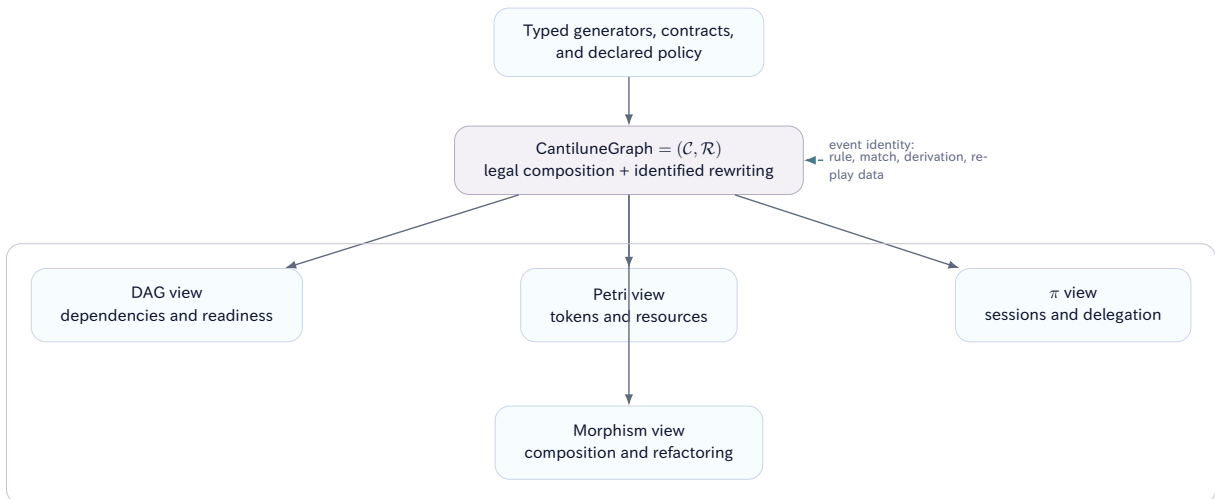
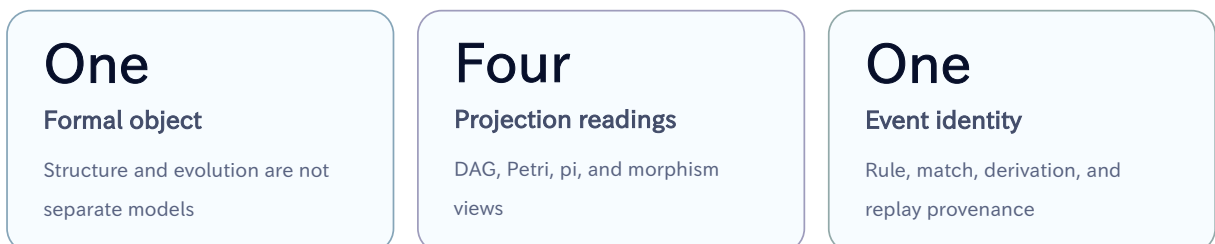


Figure / 图 2.1: Cantilune's intended unified semantic object. The core specifies legal typed composition and identified rewriting; DAG, Petri, π , and morphism are readings of that one object. Dashed cyan denotes the certificate obligation rather than a claim that every projection is automatically complete.

Source: Aligned to the formal specification and RFC-0001 (Moonweave AI, 2026g,d).

2.2 Design invariants / 设计不变量



1. **Composition is typed.** A wire connects compatible output and input ports; it is a contract and a typing obligation, not an untyped arrow.
2. **Reconfiguration is explicit.** A topology, authority, or session change is proposed as a rule application with stated admission conditions, rather than a hidden mutation.
3. **Views must earn agreement.** A non-identity projection requires a specified target transition system, event lift, preservation/reflection evidence, and terminal-observation conditions; monoidal structure alone is not enough.
4. **Model work is local to a node.** The architectural proposal is: the model proposes within a node; the formal structure and declarative policy dispose between nodes (Moonweave AI, 2026d).

2.3 The four readings / 四个读法

Table / 表 2.1: The four views are readings of the same intended object, not four product modules.

View	Question it answers	Formal obligation
DAG	What depends on what; what is runnable, blocked, or cyclic?	A certified observable transition reading for dependency progress
Petri	Which individual token, quota, lock, budget, or approval is held?	Ordered-interface, provenance-preserving resource and firing semantics
π	Which session exists; who delegated, accepted, rejected, or transferred control?	Typed/open process bridge and native late- π event correspondence
Morphism	How do pieces compose, run in parallel, and admit refactoring?	Identity reading of the symmetric monoidal structure

Interpretation boundary

The diagram and table describe a formal target. They do not assert that an LLM, tool adapter, or production control plane has already implemented these readings. The specification itself excludes runtime, LLM, I/O, and tool/network integration from the pure formal core.

Mathematical foundations and theoretical argument

数学基础与理论论证

PROOF-THEORETIC RECONSTRUCTION / 证明论重构

本章从协调世界的顶层接口出发，将 Cantilune 的形式理论重构为一个闭合的“变量-可接受性公理-定理-理论-结论”体系。正文中的每个符号均在首次使用前给出类型，每项结论均注明其成立所需的前提；经典数学来源、本文新增结构与机械证明证据彼此分开，从而不再以工程阶段、计划代号或代码文件名代替数学论证。

3.1 Top-level Modeling / 顶层建模

本节给出 Cantilune 对 Agent Orchestration 的顶层数学建模。其将协作理解为一个具有明确结构、资源归属、证据与策略状态的协调世界。该世界在每一时刻表现为配置 $X_i \in \text{Config}(\Sigma)$ ，并通过可识别、可审计且可重放的协调事件 e 演化为后继配置 X_{i+1} 。为说明何种协作结构能够被构造、何种结构变化能够发生，本节进一步从签名 Σ 引入协调演算 $\text{Calc}_\Sigma = (\mathcal{C}_\Sigma, \mathcal{R}_\Sigma)$ ：其中 $\mathcal{C}_\Sigma$ 给出类型正确的静态协作结构语言， $\mathcal{R}_\Sigma$ 给出其允许的重写动力学。由此，状态、事件、结构与演化被置于同一形式框架中，并为后续的 DPO 重写、四投影、replay 与终态分析建立共同的数学起点。

3.1.1 Coordinative World State and Events / 协调世界与协调事件

Agent Orchestration 通常被建模为多个 agent 之间互相通信，但是对于真实协作而言，orchestration 不止互通这么简单，其还需要考虑任务归属、执行权限与预算、结果可预期与可重放等等。

因此，Cantilune 将 Agent Orchestration 建模为“一个不断被合法改变的协作世界”。该世界每时刻都具有一个状态，被定义为“协调世界状态 (X)”。在给定当前签名 Σ 时， X 是一个合法运行时配置：

$$X \in \text{Config}(\Sigma). \quad (3.1)$$

它包含当前状态下的参与者、工作对象、工作关系、稀缺资源、收集到的证据、适用策略等等。该状态会不断发生变化，变化由事件产生，该事件被定义为“协调事件 (e)”。因此，整个 orchestration 的过程可以被定义为：

$$X \xrightarrow{e} X'. \quad (3.2)$$

Example / 案例

让 planner 将任务 T 委派给 coder，它是一个事件 e_{delegate} ，使世界从 X 变成 X' 。在 X 中，任务 T 由 planner 负责；在 X' 中，任务 T 由 coder 负责或由 coder 执行。

若任务需要唯一写权限，则该权限从 planner 转移给 coder；planner 与 coder 之间可能产生一个新的私有会话；reviewer 的依赖 状态变成“等待 coder 的候选结果”；事件记录里保留任务 T 、双方身份、权限 token、会话名和授权证据。

这就是 Cantilune 的顶层建模：**orchestration** 是对协作世界的受约束、可识别、可重放的状态变换。

3.1.1.1 Composition of Coordinative World / 协调世界的构成

我们用 Σ 定义协调世界中允许出现的类型，作为“协作语言与类型合同”，公式表示为：

$$\Sigma = (\text{Obj}, \text{Gen}, \text{in}, \text{out}, \text{mode}, \text{contract}) \quad (3.3)$$

其中：

1. Obj: 可出现的东西的类别，例如 Task、Artifact，以及写权限、会话和审核类型。
2. Gen: 允许出现的操作类别，例如 delegate、accept、reject、createSession。
3. in 与 out: 每一种操作需要什么、产生什么。
4. contract: 端口或操作需要满足的合同。
5. mode: 是签名中每种类型的结构使用方式，即该类型的值能否被复制、能否被丢弃,回答的不是“端口是输入还是输出”，而是一份这种类型的东西，是否能被复制，是否能被不用就扔掉。

进一步的，我们定义 $\text{Config}(\Sigma)$ ：

$$\begin{aligned} \text{Config}(\Sigma) := \Big\{ & (v, V, E, \ell, D, U, N, o_D, o_U, o_N, O, \pi, Z) \mid \\ & v \in \mathbb{N}, V \in \mathcal{P}_{\text{fin}}(\text{NodeId}), E \in \mathcal{P}_{\text{fin}}(\text{NodeId} \times \text{NodeId}), \\ & \ell \in (\text{NodeId} \rightarrow \text{Gen}(\Sigma)), D \in \mathcal{P}_{\text{fin}}(\text{DataId}), \\ & U \in \mathcal{P}_{\text{fin}}(\text{ResourceId}), N \in \mathcal{P}_{\text{fin}}(\text{NameId}), \\ & o_D \in (\text{DataId} \rightarrow \text{NodeId}), \\ & o_U \in (\text{ResourceId} \rightarrow \text{NodeId}), \\ & o_N \in (\text{NameId} \rightarrow \text{NodeId}), \\ & O \in \text{Observation}^*, \quad \pi \in \text{PolicyState}, \\ & Z \in \mathcal{P}_{\text{fin}}(\text{TombstoneId}) \Big\}. \end{aligned} \quad (3.4)$$

进而 X 可以表示为：

$$X = (v, V, E, \ell, D, U, N, o_D, o_U, o_N, O, \pi, Z) \in \text{Config}(\Sigma) \quad (3.5)$$

Table / 表 3.1: 协调世界配置的字段说明。

字段	类型	形式含义	在 Agent Orchestration 中的含义
v	$\mathbb{N}$	配置记录的签名版本	当前世界所属的版本或 epoch 标识。它用于审计与 replay；它不是签名 Σ 本身。
V	$\mathcal{P}_{\text{fin}}(\text{NodeId})$	活跃节点的有限集合	当前存在的 Agent、任务、工具调用、产物处理节点、审核节点等实例。
E	$\mathcal{P}_{\text{fin}}(\text{NodeId} \times \text{NodeId})$	活跃节点间的有限有向边集合	当前的依赖、控制、数据流或协作关系。例如“Reviewer 等待 Coder 的候选结果”。
ℓ	$\text{NodeId} \rightarrow \text{Gen}(\Sigma)$	从节点标识到签名生成元的部分标注	指出某个节点在当前运行中是什么合法操作实例，例如 Planner、Coder、delegate、review 或 createSession。
D	$\mathcal{P}_{\text{fin}}(\text{DataId})$	活跃数据 token 的有限集合	任务描述、计划、候选产物、测试结果、审核结论等可传递或可消费的工作对象。
U	$\mathcal{P}_{\text{fin}}(\text{ResourceId})$	活跃资源 token 的有限集合	不可随意复制的写权限、预算、配额、独占工具槽、锁等线性能力。
N	$\mathcal{P}_{\text{fin}}(\text{NameId})$	活跃名字或会话的有限集合	私有 channel、任务 session、临时关联标识、可被传递的通信端点。
o_D	$\text{DataId} \rightarrow \text{NodeId}$	数据 token 的归属映射	哪个活跃节点当前负责、持有或有权提交某份产物。
o_U	$\text{ResourceId} \rightarrow \text{NodeId}$	资源 token 的归属映射	谁当前持有唯一写权限、预算或独占执行资格。它保证资源转移不是悄悄复制。
o_N	$\text{NameId} \rightarrow \text{NodeId}$	名字或会话的归属映射	哪个节点当前控制、创建或负责关闭某个私有会话。
O	Observation^*	外部观察构成的有限序列	人类批准、评审意见、工具回执、环境输入、投票等。使用序列而非集合，因此发生顺序保留。
π	PolicyState	策略状态	当前授权模式、投票或 quorum 进度、重试策略、反馈聚合、epoch 内排名信息等。
Z	$\mathcal{P}_{\text{fin}}(\text{TombstoneId})$	已删除实体的墓碑集合	被删除的节点、资源或会话不会从审计历史中“失忆”；墓碑支持 trace 对账与 replay。

3.1.1.2 Composition of Coordinative Events / 协调事件的构成

与协调世界类似，我们也可以用一组变量来定义协调事件：

$$e = (v, \rho, X, X', m, \kappa, \nu, \alpha, \omega, k). \quad (3.6)$$

Table / 表 3.2: 协调事件的字段说明。

字段	回答的问题
v	此次事件发生在哪个版本或 epoch?
ρ	使用了哪条规则? 例如委派。
X	事件发生前，世界是什么状态?
X'	事件发生后，世界是什么状态?

Continued on next page

Table / 表 3.2: 协调事件的字段说明。(Continued)

字段	回答的问题
m	规则具体匹配到了 X 中的哪些实体?
κ	改变局部结构时, 哪些外部连接被保留?
ν	是否生成了新鲜名字, 例如新私有 session?
α	什么授权、策略或资源证据允许这次事件?
ω	有什么外部观察或批准证据?
k	这是内部、外部还是行政性事件?

其中尤其要注意的是 e 和 ρ 的区别, 前者是一次具体发生的事件, 后者则是“规则”, 代表一种可发生变化的模板; 同一条规则可以被多次、在不同对象上使用。

Example / 案例

假设当前配置中有: 任务 T ; Planner p ; Coder c ; 唯一写权限 w ; 尚不存在的私有会话。规则只说:

$$\rho_{\text{delegate}} = \text{“允许委派任务”}. \quad (3.7)$$

实际事件则会说:

$$e_1 = (v, \rho_{\text{delegate}}, X, X', m_1, \kappa_1, \{s\}, \alpha_1, \omega_1, k). \quad (3.8)$$

其中:

- m_1 指明: 本次匹配的是任务 T 、Planner p 、Coder c 与权限 w 。
- s 指明: 本次创建的新私有会话叫 s 。
- α_1 指明: Planner 确实有权把该任务和权限交给 Coder
- ω_1 指明: 若需要人工批准或评审记录, 它是什么。
- X' 指明: 任务责任、写权限归属、会话集合和 Reviewer 的等待关系具体变成什么。

如果 Planner 把另一个任务 T' 委派给另一个 Coder c' , 即使仍使用同一条规则, 也已经是另一件事: $e_2 \neq e_1$, 因为 $m_2 \neq m_1$ 。

3.1.2 Coordinative Calculus / 协调演算

在之前的部分, 我们定义了 $X \in \text{Config}(\Sigma)$ 作为协调世界的运行时状态, 并以协调事件 e 表示该世界的一次具体变化。然而, 仅有状态与事件记录还不足以区分: 哪些协调结构在类型上能够被构造, 以及哪些状态变化属于系统允许的协调行为。

为此, Cantilune 在签名 Σ 下引入协调演算:

$$\text{Calc}_\Sigma = (\mathcal{C}_\Sigma, \mathcal{R}_\Sigma). \quad (3.9)$$

其中, $\mathcal{C}_\Sigma$ 是由 Σ 生成的静态协调结构语言; $\mathcal{R}_\Sigma$ 是在同一签名下声明的协调变化模板集合。在接口层面, $\mathcal{C}_\Sigma$ 的对象是对象类型的有限有序组合:

$$\text{Ob}(\mathcal{C}_\Sigma) = \text{Obj}^*, \quad (3.10)$$

每个生成元 $g \in \text{Gen}$ 给出基本操作：

$$g : \text{in}(g) \longrightarrow \text{out}(g). \quad (3.11)$$

这些基本操作可通过恒等、顺序组合、并列组合、显式交换，以及 mode 所允许的复制和丢弃操作构成更大的静态协调图式。

每个 $\rho \in \mathcal{R}_\Sigma$ 是可重复使用的协调变化模板。当 ρ 在具体配置 X_i 上形成一次满足资源、授权、策略与外部证据条件的具体事件 e 时，产生运行时变化：

$$\boxed{X_i \xrightarrow{e} X_{i+1}.} \quad (3.12)$$

本节只给出协调演算的顶层接口。 $\mathcal{C}_\Sigma$ 的精确商构造、执行语义 Exec_Σ 、认证投影语义 Proj_Σ ，以及三者在签名扩展下的相容性，将在下一章中给出。

3.2 Proof Architecture and Conceptual Lineage / 证明体系与概念谱系

上一节给出了状态 X 、事件 e 、签名 Σ 与协调演算 Calc_Σ 的顶层接口。以下不再引入另一套模型，而是逐层回答五个问题：哪些量是自由变化的变量；哪些条件界定一个可接受的 Cantilune 模型；这些条件推出哪些定理；各定理如何组成一个统一理论；以及该理论最终允许作出什么结论。

Chapter thesis / 本章核心命题

固定一个满足显式可接受性条件的有限协调签名及其执行模型。若同一业务发生同时携带完整 DPO replay、四个独立语义视图的全迁移系统投影、严格签名准入、名称传递与域语义对齐证书，则该发生在 DAG、Petri、强迟 π 与态射四个视图中具有同一条可追溯语义；若再给出有限高度证据、稳定公平窗口和正进展概率下界，则执行几乎必然进入并永久留在稳定区。前一结论是条件化的相容性定理，后一结论是条件化的随机稳定化定理。

3.2.1 The five-part logical object / 五元逻辑对象

全文的数学论证组织为

$$\boxed{\mathfrak{T}_{\text{Can}} = (\mathcal{V}, \mathcal{A}, \mathcal{T}, \mathcal{M}, \mathcal{C}),} \quad (3.13)$$

其中 $\mathcal{V}$ 是全局变量与类型， $\mathcal{A}$ 是模型可接受性公理， $\mathcal{T}$ 是由这些前提机械证明的定理族， $\mathcal{M}$ 是把定理族连接为同一协调语义的组合理论， $\mathcal{C}$ 是最终相容性与稳定化结论。这里的“公理”是待研究模型的输入条件，并不是向 Lean 证明内核添加的项目自定义公理。形式化工程本身只使用基础逻辑与商类型所需的标准原则；项目定理的机械证明状态由证明清单单独记录 (Moonweave AI, 2026c,b)。

五部分之间的依赖方向是

$$\boxed{\mathcal{V} \longrightarrow \mathcal{A} \longrightarrow \mathcal{T} \longrightarrow \mathcal{M} \longrightarrow \mathcal{C}.} \quad (3.14)$$

箭头不可逆：结论不能反过来补造缺失的投影证书，代码中的一个参考实例也不能替代 对任意生产实例的可接受性验证。

3.2.2 Conceptual lineage / 概念谱系

本理论有意区分“经典结构”与“Cantilune 特化”。前者提供数学语言，后者给出为 可审计 Agent coordination 所需的加强条件。两者的边界如下。

Table / 表 3.3: 数学概念的来源与 Cantilune 特化。

层次	经典数学来源	Cantilune 中的特化
静态组合	对称幺半范畴的恒等、复合、张量与对称结构 (Mac Lane, 1998)	有限有类型签名、结构使用 mode，以及只在 mode 许可时出现的显式 copy/drop primitive。本文不额外假定余幺半群方程。
结构重写	粘合范畴中的双推前重写与并行独立 (Lack and Sobociński, 2004, 2005)	有限精确位置编码的有类型开放超图、边界保持、新鲜支撑与可执行代表元。
并发资源	Petri net 的函子语义与 firing 解释 (Bruni et al., 2001)	对同一选中 occurrence 的声明增量、incidence、enabling、firing 与 token provenance 等式。
名称传递	强迟 π 语义及其世界索引指称模型 (Fiore et al., 2002)	request/accept 子语言、15 类审计事件族、稳定元数据、原生强迁移与四视图对齐。
递归与非确定性	ω CPO、连续映射与递归域方程的 domain theory (Abramsky and Jung, 1994)	下 ω -Scott 闭集幂结构、名称抽象、隐藏、有限 Hoare 与有限 trace 观察边界。
随机稳定化	有限 Markov kernel、停时与几何尾界	有限高度证据、上闭稳定区、epochwise fairness、原生事件标签和精确 replay。

因此，本章不会把“用了某个经典概念”写成“经典文献已经证明了 Cantilune 的结论”。经典结果只说明数学工具的来源；所有跨视图、审计、准入和 replay 结论都仍需本文的显式证书。

3.2.3 Dependency spine / 依赖主干

全文唯一的定义依赖主干为

$$\begin{aligned}
 \Sigma &\longrightarrow \mathcal{C}_\Sigma \longrightarrow \text{Hyp}_\Sigma \longrightarrow \text{DPO}_\Sigma \longrightarrow \text{Event}_\Sigma \longrightarrow \mathcal{L}_\Sigma \\
 &\longrightarrow (\mathcal{L}_{\text{DAG}}, \mathcal{L}_{\text{Petri}}, \mathcal{L}_\pi, \mathcal{L}_{\text{Mor}}) \longrightarrow \text{FourCoh} \\
 &\searrow \text{Agent} \qquad \searrow (Q, U, K) \longrightarrow \text{Stable}.
 \end{aligned} \tag{3.15}$$

Hyp_Σ 表示有类型开放超图范畴； DPO_Σ 表示其 DPO 动力学； $\mathcal{L}_\Sigma$ 是源执行 LTS；括号内恰为四个语义投影；Agent 是 π 行的指称解释，并不是第五个投影； (Q, U, K) 则是反馈层的证据空间、稳定区与 Markov kernel。

3.3 Global Variables and Admissibility Axioms / 全局变量与可接受性公理

3.3.1 Typed variable ledger / 有类型变量表

固定下列变量。除非另有说明，本章中的所有定理都在这些类型下陈述。

Table / 表 3.4: 全局数学变量。

变量	类型	含义
Σ	有限签名 ($O, G, \text{in}, \text{out}, \text{mode}, \text{contract}$)	原子对象、primitive、端口词、结构使用权限与合同。
w, u, v	O^*	原子对象组成的有限有序词，即静态接口。
g	G	一个 primitive，具有 $g : \text{in}(g) \rightarrow \text{out}(g)$ 。
X, Y, Z	$\text{Config}(\Sigma)$	源执行中的协调世界状态。
ρ	$\mathcal{R}_\Sigma$	规则 span；它是可重复使用的变化模板。
m	合法 match	将规则左侧单态地嵌入具体状态。
e	Event_Σ	携带版本、规则、match、接口、新鲜性与证据的一次 occurrence。
r	Recipe_Σ	不存储终点的确定性 replay recipe。
$\mathcal{L}$	$(S, E, \sim, \rightarrow, \text{obs}, \text{succ}, \text{wait}, \text{ver})$	可观察 LTS； $\sim$ 是状态等价关系。
α	$\Sigma \hookrightarrow \Sigma^+$	保持类型词、mode 与 contract 的签名扩展。
j	$\{\text{D}, \text{N}, \pi, \text{M}\}$	DAG、Petri、强迟 π 、态射四个视图的索引。
P_j	$\mathcal{L}_\Sigma \Rightarrow \mathcal{L}_j$	从源 LTS 到第 j 个独立目标 LTS 的投影证书。
n, m	$\mathbb{N}$	名称世界的有限基数。
ι	$n \hookrightarrow m$	世界之间的有限注入。
A	$\text{World} \rightarrow \omega\text{CPO}$	世界索引的 Agent 域。
Q	有限高度 join 半格	反馈证据空间。
U	$U \subseteq Q$	向上闭的稳定区。
K	$S \times S \rightarrow [0, 1]$	有限状态 Markov kernel。
H	$\mathbb{N}$	证据高度的统一上界。
ε	$(0, 1]$	每次公平机会的进展概率下界。

四视图记号只用于区分数学语义，不表示四份互不相关的数据。最终结论要求它们通过 同一个 e 、同一个版本与同一个 occurrence identity 连接。

3.3.2 Admissibility axioms / 可接受性公理

下列十项公理定义“完整可接受模型”。它们不是无条件的世界规律，而是总定理的精确 输入接口。任何缺失项都会使相应结论停止，而不会由其余项自动补全。

- A1. Finite typed signature / 有限有类型签名。** O 与 G 有限且等价可判定；每个 g 的输入输出词有类型；copy 或 drop 只在 mode 给出相应许可证时可形成。合同随合法构造保持。
- A2. Adhesive DPO legality / 粘合 DPO 合法性。** 规则腿与 match 满足单态、gluing、边界保持和新鲜支撑条件；所需推前补与第二推前 均存在。并发结论还要求两次直接导出具有显式 parallel-independence 分解。
- A3. Complete event and deterministic replay / 完整事件与确定性重放。** 每个被认证事件都携带足以

重建 DPO 结果的 recipe; recipe 不把目标状态作为输入, replay kernel 对给定 recipe 和源状态是确定的。

- A4. Quotient-compatible execution / 商相容执行。** 若 $X \sim X_1$ 、 $Y \sim Y_1$ 且 $X \xrightarrow{e} Y$, 则存在对应迁移 $X_1 \xrightarrow{e_1} Y_1$ 。观察、成功、等待与版本在 $\sim$ 下良定义。
- A5. Complete four-view projections / 完整四视图投影。** 对每个 $j \in \{D, N, \pi, M\}$, P_j 同时给出状态映射、事件映射、soundness、从投影像出发的每步 reflection、事件 Lift, 以及终态、版本和资源保持; 证书覆盖整个目标 LTS, 而不只是一条被选中边。
- A6. Extension naturality and strict admission / 扩展自然性与严格准入。** $\alpha : \Sigma \hookrightarrow \Sigma^+$ 在四视图中自然重索引; 新规则的加入由独立的 strict-admission 见证承担。单纯 reindex 不得冒充准入。
- A7. Selected occurrence adequacy / 选中发生的充分证据。** 被选中的 $X \xrightarrow{e_*} Y$ 携带原生迁移、精确 replay、资源变化、静止性或 rank、qualification、authorization 与 occurrence identity。此处的业务 rank 与后文 SCC rank、反馈高度均为不同变量。
- A8. Occurrence sidecars / 发生级伴随证书。** DAG 凝聚与 Petri firing 对同一 e_* 给出局部 sidecar; Petri sidecar 精确陈述 声明增量、incidence、enabling、firing 和 token provenance, DAG sidecar 精确陈述 SCC 内部边或凝聚边的二分。
- A9. Name-passing and denotational seam / 名称传递与指称接缝。** 选中 registry row 的事件坐标、稳定元数据、原生延迟 π 导数以及 FMS Agent 端点字面相等; “大致对应”或仅存在弱迁移不足以满足本公理。
- A10. Finite-height stochastic progress / 有限高度随机进展。** Q 有高度界 H , U 向上闭, 正概率 kernel 边均有原生事件标签; 存在稳定 epoch 窗口、epochwise fairness、单调证据与每次机会至少 ε 的严格进展概率, 并且事件与 epoch 对齐。

Logical boundary / 逻辑边界

A5 是全 LTS 证书, A8 是选中 occurrence 的局部语义说明; 二者不能互换。A9 把 FMS 绑定到 π 行, 但不会产生第五个投影。A10 是有条件的公平进展假设, 因此后文的概率 1 结论不是无条件 liveness。

3.4 Static Coordination Syntax / 静态协调语法

3.4.1 Raw diagrams and the quotient category / 原始图式与商范畴

令 O^* 为 O 上的有限词。对 $u, v \in O^*$, 原始有类型图式 $d : u \Rightarrow_{\Sigma} v$ 由下列构造归纳生成:

$$\begin{aligned} d ::= & \text{id}_u \mid \text{gen}(g) \mid d_2 \circ d_1 \mid d_1 \otimes d_2 \mid \sigma_{u,v} \\ & \mid \text{copy}_a(p) \mid \text{drop}_a(q), \end{aligned} \quad (3.16)$$

其中 $g \in G$, p 与 q 分别是 $\text{mode}(a)$ 允许复制与丢弃的证据。因此 copy/drop 是带许可证的显式 primitive。仅凭 $\text{mode}(a) = \text{Cartesian}$ 或 Affine , 本理论并不额外加入 自然交换余么半群律。

令 $\equiv_{\text{smc}}$ 是由范畴律、张量双函子律、结合子与么元子自然性、对称的互逆性, 以及 pentagon、triangle、hexagon 相容性生成的最小同余。定义

$$\mathcal{C}_{\Sigma}(u, v) = \{d : u \Rightarrow_{\Sigma} v\} / \equiv_{\text{smc}}. \quad (3.17)$$

复合与张量在该商上良定义, 故 $\mathcal{C}_{\Sigma}$ 是由 Σ 生成、并配备 mode-admissible distinguished copy/drop

generators 的自由对称幺半范畴。

Theorem: free interpretation / 定理：自由解释

设 $\mathcal{D}$ 为对称幺半范畴。若为每个 $a \in O$ 选择目标对象，为每个 $g \in G$ 选择类型正确的目标态射，并为许可的 copy/drop 选择与 primitive 解释相容的态射，则存在一个保持这些选择的对称幺半函子

$$[[-]]: \mathcal{C}_\Sigma \longrightarrow \mathcal{D}.$$

任意两个在原子对象与全部 primitive 上一致的此类解释之间，存在唯一具有给定原子分量的幺半自然同构。

该定理说明：静态协调图式的意义由原子类型和 primitive 的意义唯一控制；它没有把运行时 DPO 事件、概率选择或外部批准塞入静态语法。动态语义必须在下一层另行给出。

3.4.2 Extension of syntax / 语法扩展

签名扩展

$$\alpha: \Sigma \hookrightarrow \Sigma^+ \quad (3.18)$$

由对象与 primitive 的单射组成，并严格保持输入词、输出词、mode 与 contract。逐构造重索引得到

$$\alpha_C: \mathcal{C}_\Sigma \longrightarrow \mathcal{C}_{\Sigma^+}. \quad (3.19)$$

Theorem: coherent signature extension / 定理：相容签名扩展

若 $\Sigma \xrightarrow{\alpha} \Sigma' \xrightarrow{\beta} \Sigma''$ ，则

$$(\beta \circ \alpha)_C = \beta_C \circ \alpha_C, \quad (\text{id}_\Sigma)_C = \text{id}_{\mathcal{C}_\Sigma}.$$

同一相容性同时保持对象词、primitive、mode 与 contract，并可被四个语义视图继承。

重索引只把旧结构放入更大签名；它不证明新 primitive 已获授权。真正改变允许规则集合的动作称为 *strict admission*，必须另有授权、资格与版本证书。这个区分将在跨 epoch 理论中再次使用。

3.5 Typed Open-Hypergraph Dynamics / 有类型开放超图动力学

3.5.1 The ambient adhesive category / 环境粘合范畴

把 Σ 的对象与 primitive 编码为有限有类型开放超图。其节点与超边带类型，输入输出边界保留位置顺序，因而静态词接口可以被精确恢复。记所得范畴为 Hyp_Σ 。它等价于一个有类型预层 slice 中的有限、精确、可重现全子范畴；后者为粘合范畴，故沿单态的推前具有 DPO 所需的 Van Kampen 性质 (Lack and Sobociński, 2004, 2005)。

位置编码桥只在上述有限精确范围内使用。它不声称任意无限 slice 对象都能由运行时开放超图表示。

3.5.2 Rules, matches and direct derivations / 规则、匹配与直接导出

一条规则是 Hyp_Σ 中的单态 span

$$\rho = \left(L \xleftarrow{\ell} K \xrightarrow{r} R \right). \quad (3.20)$$

L 是待替换局部图, K 是必须保留的接口, R 是替换结果。对单态 match $m : L \hookrightarrow X$, 直接导出由两个推前方块给出:

$$\begin{array}{ccc} K & \xrightarrow{\ell} & L \\ \downarrow k & & \downarrow m \\ D & \xrightarrow{d} & X \end{array} \quad \begin{array}{ccc} K & \xrightarrow{r} & R \\ \downarrow k & & \downarrow n \\ D & \xrightarrow{h} & Y. \end{array} \quad (3.21)$$

左方块给出 pushout complement D , 右方块给出重写结果 Y 。开放边界、未删除接口、资源归属与新鲜名字均在 match 合法性中检查, 而不是在重写后补救。

Theorem: two levels of DPO uniqueness / 定理: DPO 的两层唯一性

对同一 complement span 和同一规则右腿, 任意两个第二推前结果 Y_1, Y_2 之间存在 唯一保持余锥腿的典范同构

$$\varphi : Y_1 \cong Y_2.$$

进一步, 在有限精确表示中固定 inclusion match 与具体 fresh support 后, 计算构造选出 典范代表元; 满足同一计算规范的两个认证端点字面相等, 且 proof representative 的选择 不改变该端点。

第一层是范畴论中的“唯一到同构”, 第二层才是执行器中的“同一确定端点”。这两层共同支持 replay, 但不可压缩成笼统的“DPO 结果绝对唯一”。

3.5.3 Parallel independence / 并行独立

考虑同一 X 上的两次直接导出

$$X \xrightarrow{\rho_1, m_1} Y_1, \quad X \xrightarrow{\rho_2, m_2} Y_2. \quad (3.22)$$

若 m_1, m_2 给出显式 parallel-independence 分解, 即任一规则删除或创建的部分不破坏 另一规则所需的保留部分, 则两步均存在 residual:

$$Y_1 \xrightarrow{\rho_2/m_1} Z_{12}, \quad Y_2 \xrightarrow{\rho_1/m_2} Z_{21}. \quad (3.23)$$

Theorem: conditional concurrency diamond / 定理: 条件并发菱形

在 A2 的粘合性与 parallel-independence 前提下, 存在保持两条规则右腿的 典范同构

$$Z_{12} \cong Z_{21}.$$

因而两次独立变化的执行顺序不改变最终协调结构，至多改变其表示。

这是有条件的交换菱形，不是全局 confluence、Church–Rosser 或任意事件可交换性。具有资源冲突、名称捕获或共同删除对象的 match 不在结论范围内。

3.6 Observable Execution and Replay / 可观察执行与重放

3.6.1 Complete events / 完整事件

顶层事件元组中的 X' 适合审计，但不能被当作 replay 的计算依据，否则“重放成功”会退化为直接读取已存终点。为此，从完整事件 e 中删除源、终点和任何证明对象，得到 recipe

$$r(e) = (v, \rho, m, \kappa, \nu, \alpha, \omega, k, \dots) \in \text{Recipe}_\Sigma, \quad (3.24)$$

其中省略号只表示完成具体有限 DPO 构造所需的确定数据，不包含目标状态。replay kernel 是偏函数

$$\text{run}_\Sigma : \text{Recipe}_\Sigma \times \text{Config}(\Sigma) \rightarrow \text{Config}(\Sigma). \quad (3.25)$$

其定义域恰由版本一致、match 合法、边界保持、新鲜性、合同与授权检查确定。

Theorem: replay uniqueness / 定理：重放唯一性

若同一 recipe r 在同一源状态 X 上得到两个认证结果，

$$\text{run}_\Sigma(r, X) = Y_1, \quad \text{run}_\Sigma(r, X) = Y_2,$$

则 $Y_1 = Y_2$ 。因此认证事件的目标由源和 endpoint-free recipe 唯一决定，而不是由被记录的目标、rule identifier 或非确定的 proof representative 决定。

该结论依赖前节 DPO 的两层唯一性：范畴层排除语义歧义，有限精确表示层选出字面相等 的可执行端点。

3.6.2 Observable quotient LTS / 可观察商迁移系统

一个可观察 LTS 定义为

$$\mathcal{L} = (S, E, \sim, \rightarrow, \text{obs}, \text{succ}, \text{wait}, \text{ver}), \quad (3.26)$$

其中 $\sim$ 是 S 上的等价关系， $\rightarrow \subseteq S \times E \times S$ ；obs 给出外部观察，succ 与 wait 是状态谓词， $\text{ver} : S \rightarrow \mathbb{N}$ 给出版本。定义

$$\text{Normal}(x) \iff \nexists e, y, (x \xrightarrow{e} y \wedge \text{obs}(e)). \quad (3.27)$$

一般 LTS 的三个正常终态为

$$\begin{aligned}
\text{Success}(x) &\iff \text{Normal}(x) \wedge \text{succ}(x), \\
\text{ExternalWait}(x) &\iff \text{Normal}(x) \wedge \neg \text{succ}(x) \wedge \text{wait}(x), \\
\text{Deadlock}(x) &\iff \text{Normal}(x) \wedge \neg \text{succ}(x) \wedge \neg \text{wait}(x).
\end{aligned} \tag{3.28}$$

Theorem: quotient well-definedness and terminal partition / 定理：商良定义与终态三分

在 **A4** 下，迁移、观察、成功、等待与版本均下降到商集 $S/\sim$ 。每个 normal 状态恰属于 Success、ExternalWait、Deadlock 三类之一，且三类两两不交。

生产性的无限执行不是第四种 normal 状态，因为它仍可继续迁移。后文针对某个具体 admitted occurrence 的“四类执行分类”是在路径层加入 stable productive infinity，不能与式 (3.28) 混同。

3.6.3 Paths and replayable traces / 路径与可重放轨迹

有限路径递归定义为

$$\begin{aligned}
&\text{Path}_{\mathcal{L}}(x, x) \ni \epsilon, \\
&p \in \text{Path}_{\mathcal{L}}(x, y), y \xrightarrow{e} z \implies p \cdot e \in \text{Path}_{\mathcal{L}}(x, z).
\end{aligned} \tag{3.29}$$

若路径中的每条边都携带 recipe 且逐步满足 $\text{run}(r(e_i), X_i) = X_{i+1}$ ，则称其为 replayable trace。于是执行证据同时具有两个互补面向：LTS 路径给出行为组合，DPO recipe 给出每一步的构造性重现。

3.7 Signature Growth and Epoch Semantics / 签名增长与 Epoch 语义

3.7.1 Reindexing versus admission / 重索引与准入

设

$$\Sigma_0 \xrightarrow{\alpha_0} \Sigma_1 \xrightarrow{\alpha_1} \dots \xrightarrow{\alpha_{t-1}} \Sigma_t \tag{3.30}$$

为有限扩展链。每个 α_i 诱导配置、事件、静态图式与四视图状态的重索引。纯重索引保持旧版本事实；它既不增加新规则，也不改变一条旧事件的授权来源。

若 $\rho_{\text{new}} \in G_{i+1} \setminus \alpha_i(G_i)$ ，则严格准入是额外对象

$$\text{Adm}_i(\rho_{\text{new}}) = (\text{qualified}, \text{authorized}, \text{quiescent}, \text{resourceSafe}, \text{versionStep}). \tag{3.31}$$

只有 Adm_i 才允许把新规则用于 Σ_{i+1} 的原生事件。

Theorem: finite dependent epoch coherence / 定理：有限依赖 Epoch 相容性

对任意链 (3.30)，复合重索引与逐步重索引相同；若每次真正扩张都携带严格准入证书，则源路径、事件 recipe、四视图路径、版本与 admission marks 可沿整个链共同运输。运输后的每步仍对应原 occurrence，而不是一个仅共享 rule identifier 的新事件。

3.7.2 Version invariant / 版本不变量

对事件 $e : X \rightarrow Y$ ，定义版本合法性

$$\text{EpochOK}(e) \iff \text{ver}(X) = \text{ver}(e) = \text{ver}(Y), \quad (3.32)$$

行政准入事件可把下一状态推进至新版本，但该推进必须由 `versionStep` 明示。于是“结构被重索引到新签名”和“世界已承认新规则”在数学上是两种不同的边。

3.8 Certified Semantic Projections / 认证语义投影

3.8.1 Projection certificate / 投影证书

设源与目标分别为 $\mathcal{L}_S = (S_S, E_S, \sim_S, \rightarrow_S, \dots)$ 和 $\mathcal{L}_T = (S_T, E_T, \sim_T, \rightarrow_T, \dots)$ 。投影证书

$$P = (f_S, f_E, \text{Lift}, \text{sound}, \text{reflect}, \text{terminal}, \text{version}, \text{resource}) \quad (3.33)$$

包含状态映射 $f_S : S_S \rightarrow S_T$ 、事件映射 $f_E : E_S \rightarrow E_T$ ，并满足：

$$x \xrightarrow{e}_S y \implies f_S(x) \xrightarrow{f_E(e)}_T f_S(y), \quad (3.34)$$

$$f_S(x) \xrightarrow{a}_T z \implies \exists e, y, x \xrightarrow{e}_S y \wedge \text{Lift}(e, a) \wedge z \sim_T f_S(y). \quad (3.35)$$

此外 f_S 保持并反射三个 normal 终态，保持版本，并按照目标视图的资源表示保持 ownership/token 等式。

Theorem: path lifting and reflection / 定理：路径提升与反射

每条源路径 $p : x \rightsquigarrow_S y$ 都逐边映射为 $f_P(p) : f_S(x) \rightsquigarrow_T f_S(y)$ 。反之，从 $f_S(x)$ 出发的每条原生目标路径都可逐边 Lift 为一条源路径；目标端点与被提升源端点的投影在 $\sim_T$ 下相等。成功、外部等待、死锁与版本在两个方向上相容。

因此“投影”并非画图函数。只有 soundness 会允许目标丢失行为；加入从投影像出发的 reflection 后，目标视图才不会凭空产生源语义无法解释的边。

3.8.2 The four views / 四个视图

对同一源 LTS 固定

$$\mathbf{P} = (P_D, P_N, P_\pi, P_M), \quad (3.36)$$

分别指向 DAG、Petri、强迟 π 与态射 LTS。源加四视图形成 five-view agreement；“四投影”本身始终只指目标四元组。

3.8.2.1 DAG view / DAG 视图

任意开放超图都可能含 self-loop 或一般有向环，因此不存在对所有源对象保持 incidence 的 total strict-DAG projection。总定义采用 SCC 凝聚：

$$\text{Cond}(G) = (\text{SCC}(G), E_{\text{SCC}}), \quad (3.37)$$

每条原边要么位于同一 SCC 内部，要么诱导凝聚边。定义 $r_{\text{SCC}}(C)$ 为 SCC 偏序下的 canonical lower-cardinality rank，则

$$C \longrightarrow C' \implies r_{\text{SCC}}(C) < r_{\text{SCC}}(C'), \quad (3.38)$$

从而 $\text{Cond}(G)$ 无有向环。这个 rank 只证明凝聚 DAG 的 acyclicity，与业务规则下降量和反馈证据高度 H 均无关。

3.8.2.2 Petri view / Petri 视图

Petri LTS 把资源条件表达为 marking 与 firing。对一个被选中的准入后 occurrence e_* ，局部 sidecar 给出

$$M \xrightarrow{t_*} M', \quad M' = M - \text{pre}(t_*) + \text{post}(t_*), \quad (3.39)$$

并证明：旧 declaration 经签名扩展形成精确前缀；新规则对应一个新 transition；source incidence 与 enabling 成立；被消费和产生 token 的 provenance 与 e_* 的资源坐标相同。Petri 的经典组合思想提供语义背景 (Bruni et al., 2001)，但这些 provenance 等式属于 Cantilune 的 occurrence 证书。

3.8.2.3 Morphism view / 态射视图

态射 LTS 记录源图式或开放超图之间的结构态射，把同一 occurrence 解释为满足接口与合同的态射变化。 P_M 的价值在于保留组合边界：DAG 强调依赖序，Petri 强调资源守恒， π 强调名称传递，而态射视图证明这三种读法没有遗失 $\mathcal{C}_\Sigma$ 的类型接口。

3.8.3 Whole-LTS certificate versus occurrence sidecar / 全 LTS 证书与发生级伴随证书

Non-interchangeability / 不可互换性

P_D 、 P_N 与 P_M 是产品级全 LTS 假设，因此才能导出任意有限路径的 soundness 与 reflection。SCC 凝聚和式 (3.39) 是被选中 occurrence 的局部 sidecar，只说明该边的结构和资源含义。局部 sidecar 不能反推出全 LTS 投影，反之全 LTS 投影也不会自动给出精确 token provenance。

3.9 Name-Passing and Protocol Semantics / 名称传递与协议语义

3.9.1 Categorical boundary of open processes / 开放进程的范畴边界

令 $\text{Proc}(u, v)$ 为具有极化输入输出接口的开放进程项。原始进程语法上的 α -转换、并行交换结合幺元律与 scope laws 生成同余 $\equiv_\pi$ 。定义

$$\text{OpenPi}(u, v) = \text{Proc}(u, v) / \equiv_\pi. \quad (3.40)$$

plug 给出复合，parallel 给出张量，空进程给出幺元，名字交换给出对称；hide/restriction 在接口满足条件时闭合。

Theorem: open-process categorical and operational bridge / 定理：开放进程的范畴-操作桥
商 $\text{Hom OpenPi}(u, v)$ 满足所陈述的对称幺半范畴方程，证明携带的极化接口 在 plug、parallel 与 hide 下保持类型；这些构造同时在原生强迟 π 操作语义中闭合。

范畴结构位于商 Hom 上，原始 π 进程语法本身不是范畴。这一边界防止把 结构同余前后的代表元差异误写成业务事件差异。

3.9.2 Strong late semantics and typed erasure / 强迟语义与类型擦除

写作

$$P \xrightarrow{\mu}_{\text{late}} Q \quad (3.41)$$

表示一步原生强迟迁移，不经过弱闭包。若 $P \xrightarrow{\mu}_{\text{typed}} Q$ 是合法有类型标准迁移，类型擦除 $|\cdot|$ 满足

$$|P| \xrightarrow{|\mu|}_{\text{late}} |Q|. \quad (3.42)$$

Theorem: forward operational erasure / 定理：正向操作擦除

每个有类型标准 π 迁移擦除后都是一个真正的原生无类型强迟迁移。该结论只有 typed-to-untyped 方向；它不声称任意无类型导数都能恢复类型。

3.9.3 The request-accept calculus / 请求-接受演算

定义有限源演算 $\mathcal{L}_{\text{RA}}$ ，其核心状态为

$$q_0 = \text{Idle}, \quad q_1 = \text{Requested}, \quad q_2 = \text{Accepted}, \quad (3.43)$$

并有两条可观察业务边

$$q_0 \xrightarrow{\text{request}} q_1 \xrightarrow{\text{accept}} q_2. \quad (3.44)$$

编码 η_{RA} 把它映入未经筛选的结构强迟 π 商 LTS。

Theorem: exact request-accept certificate / 定理：精确请求-接受证书

式 (3.44) 的每一步均编码为真正的原生强迟迁移；从编码像出发的 对应原生迁移又反射出唯一源事件和结构同余端点。因此该证书是双向的强迁移证书，没有用弱迁移隐藏中间步骤。

$\mathcal{L}_{RA}$ 是一个独立的协议子语言。它当前不携带 selected product candidate 所需的 session/correlation 元数据，因此本理论不声称全局 request 事件在 因果上先于某个被选中的产品 occurrence。

3.9.4 Enriched auditable mobility / 丰富可审计移动语义

裸三元组 (P, μ, Q) 无法唯一恢复 rule、session、correlation 与 occurrence identity。故定义丰富标签

$$\hat{\mu} = (\mu, \text{family}, \text{operation}, v, \text{rule}, \text{session}, \text{correlation}, \text{occurrence}, \text{phase}), \quad (3.45)$$

并以 derivative-alpha 与结构同余联合取商。稳定身份元数据位于进程商之外，phase 也不被结构同余吞并。

规范 registry 含 15 类事件族；每类事件在四视图矩阵中各占一列，形成 $15 \times 4 = 60$ 个语义单元。另有 60 个封闭登记操作位置。后两个“60”是不同集合：前者是事件族-视图矩阵坐标，后者是 API registry 的有限索引。

Theorem: enriched strong-step completeness / 定理：丰富强步完备性

每个规范事件族及每个登记操作位置均存在原生第一步；需要细化的操作携带显式原生 phase。原始结构 LTS 与丰富商 LTS 之间的映射同时 sound 且 reflective，并保持式 (3.45) 中的稳定 identity。所有步骤都是 strong late steps。

反向边界同样是定理的一部分：不存在仅从裸 (P, μ, Q) 恢复全部 15 类业务事件及 审计身份的总函数。

3.9.5 Execution classification for an admitted occurrence / 准入发生的执行分类

对一个固定 admitted occurrence，从业务起点出发的完整执行恰落入以下四类之一：

1. 到达 Success；
2. 到达 ExternalWait；
3. 到达 Deadlock；
4. 形成稳定且持续生产观察的无限轨迹。

该四分结论由一个 iff 分类定理与相邻的两两不交定理共同给出。第四类是路径性质，前三类是式 (3.28) 的 normal 状态性质。

3.10 World-Indexed Domain Interpretation / 世界索引域解释**3.10.1 Worlds, actions and name abstraction / 世界、动作与名称抽象**

令 World 的对象为有限序数 $n \in \mathbb{N}$ ，态射为有限注入 $\iota : n \hookrightarrow m$ 。语义对象是函子

$$X : \text{World} \longrightarrow \omega\text{CPO}. \quad (3.46)$$

在世界 n 上, 名称对象记为 $N(n)$, 名称抽象满足

$$(N \Rightarrow X)(n) \cong X(n)^n \times X(n+1). \quad (3.47)$$

世界移位函子 δ 定义为

$$(\delta X)(n) := X(n+1), \quad (3.48)$$

并沿有限注入自然作用。于是动作函子为

$$(\text{HX})(n) = N(n) \times (N \Rightarrow X)(n) + N(n) \times N(n) \times X(n) + N(n) \times (\delta X)(n) + X(n). \quad (3.49)$$

四个和项分别承载 bound input、free communication、fresh-name movement 与内部延续的世界索引形状。具体标签方向由操作桥确定, 式 (3.49) 只固定递归域的类型轮廓。

3.10.2 The unseparated lower semantics / 非分离下闭语义

令 $P_{\downarrow\omega}$ 为下 ω -Scott 闭集幂结构, 并选定其对称 Fubini。实际 Agent 函子为

$$F = P_{\downarrow\omega} \circ H. \quad (3.50)$$

递归域 A 满足

$$\text{fold} : FA \cong A : \text{unfold}. \quad (3.51)$$

在连续自然映射范畴中, fold 给出初始代数, unfold 给出终余代数。restriction/hiding 是自然变换, 并满足一次 unroll 方程及相应唯一性。

Theorem: maximum-compatible domain closure / 定理: 最大相容域闭包

非分离分支在全部世界索引 ωCPO 上给出 $P_{\downarrow\omega}$ 、对称 Fubini、式 (3.51)、hiding, 以及有限 Hoare、受卫 finite trace 和 contextual-Hoare 观察下的充分性与完全抽象。有限生成观察具有相应 definability。

这里的完全抽象明确受观察语言限制。它不是 constructor-sensitive strong bisimulation 的无条件完全抽象, 也不是“每个 ωCPO 元素都由一个源语法项定义”。

3.10.3 The separated evidence branch / 分离证据分支

另一分支保留更细的源观察分离, 得到 all-source solution set 与 CPO-enriched adjunction。然而其 canonical sequential Fubini 不交换。反之, 非分离分支具有选定的对称下闭结构, 却把 nullary divergence 与 deadlock 映到同一个 effect bottom。

separated branch	unseparated lower branch	
source separation, enriched adjunction	symmetric Fubini, domain equation, hiding	(3.52)
$\neg$ canonical commutative Fubini	divergence effect = deadlock effect.	

两分支组成的是 maximum-compatible ledger，而不是一个可把全部优点合并的单一模型。操作层仍以式 (3.28) 区分 deadlock 与其他终态；非分离域模型的 effect collapse 不会删除该操作区分。

3.10.4 The exact denotational seam / 精确指称接缝

对 registry 中被选中的丰富迁移

$$\hat{P} \xrightarrow{\hat{\mu}} \hat{Q}, \quad (3.53)$$

令 $\llbracket - \rrbracket_A$ 为到 Agent 域的解释。A9 要求

$$\begin{aligned} \text{event}(\hat{\mu}) &= e_*, \\ \text{meta}(\hat{\mu}) &= (v, \text{rule}, \text{session}, \text{correlation}, \text{occurrence}), \\ \text{raw}(\hat{P} \xrightarrow{\hat{\mu}} \hat{Q}) &= (P \xrightarrow{\mu}_{\text{late}} Q), \\ \text{agentEndpoints}(\hat{\mu}) &= (\llbracket P \rrbracket_A, \llbracket Q \rrbracket_A). \end{aligned} \quad (3.54)$$

这些是字面坐标等式，而不是事后推测的相似性。FMS 因而解释 π 行的端点与观察，却不承担 DAG、Petri 或态射投影的路径 reflection。

3.11 Finite-Height Feedback and Stochastic Stabilization / 有限高度反馈与随机稳定化

3.11.1 Evidence order and stable region / 证据序与稳定区

令 $(Q, \sqsubseteq, \sqcup, \perp)$ 为有限高度 join 半格。存在 $h : Q \rightarrow \{0, \dots, H\}$ ，使严格增长蕴含

$$q \sqsubset q' \implies h(q) < h(q'). \quad (3.55)$$

令 $U \subseteq Q$ 为向上闭集：

$$q \in U, \quad q \sqsubseteq q' \implies q' \in U. \quad (3.56)$$

对源执行状态给出证据映射 $\text{ev} : S \rightarrow Q$ 。一次原生事件只允许保持或增加证据：

$$x \xrightarrow{e} y \implies \text{ev}(x) \sqsubseteq \text{ev}(y). \quad (3.57)$$

Theorem: hard forward invariant / 定理：硬前向不变量

若 $\text{ev}(x_0) \in U$ ，则从 x_0 出发的任意有限执行 $x_0 \rightsquigarrow x_n$ 都满足 $\text{ev}(x_n) \in U$ 。等价地，任意有限 join 累积一旦进入 U 就不会退出。

证明只使用式 (3.56) 与 (3.57) 的归纳，与概率选择无关。这是“最终一直稳定”中“保持”部分的确定性基础。

3.11.2 Native finite Markov kernel / 原生有限 Markov Kernel

令 S 有限， $K(x, y) \in [0, 1]$ 满足

$$\sum_{y \in S} K(x, y) = 1. \quad (3.58)$$

要求每条正概率边都有原生事件标签：

$$K(x, y) > 0 \implies \exists e, x \xrightarrow{e} y \wedge \text{run}(r(e), x) = y. \quad (3.59)$$

从初始状态 x_0 与 K 得到轨迹测度 μ_{x_0} 。稳定命中时刻为

$$T_U(\omega) = \inf\{t \in \mathbb{N} \mid \text{ev}(\omega_t) \in U\}. \quad (3.60)$$

在稳定 epoch 窗口内，若当前尚未稳定，则 epochwise fairness 提供一次机会，使证据以至少 ε 的概率严格增长。令 q_n 为连续 n 次公平机会均未产生下一次严格证据增长的概率，则几何尾界为

$$q_n \leq (1 - \varepsilon)^n. \quad (3.61)$$

式 (3.61) 首先是 opportunity count 的界。只有 opportunity 与连续 epoch 对齐时，才能把它解释为真实 epoch 计数；这正是 A10 中 epochwise fairness 与 event-epoch alignment 的作用。

Theorem: replayable stochastic stabilization / 定理：可重放随机稳定化

在 A3、A4 与 A10 下，

$$\mu_{x_0}(\text{EventuallyAlways } U) = 1, \quad \mathbb{E}_{x_0}[T_U] \leq \frac{H}{\varepsilon}.$$

被抽取的每条正概率进展路径同时携带原生事件、精确 replay 与 epoch 对齐见证；命中 U 后，硬前向不变量保证永久留在 U 。

该定理不把概率 1 解释为“每一条轨迹都稳定”，也不对缺乏稳定窗口、公平机会或统一正下界的调度器作活性保证。

3.12 The Integrated Cantilune Theory / Cantilune 组合理论

3.12.1 A complete certified model / 完整认证模型

固定一次签名增长 $\alpha : \Sigma \hookrightarrow \Sigma^+$ 、源执行 $\mathcal{L}_{\Sigma^+}$ 与候选索引 c 。不含随机调度的核心认证模型记为

$$\text{CoreCert}(c, e_*) := \left(\begin{array}{l} \text{A1-A9 的数据与证明,} \\ X \xrightarrow{e_*} Y, r(e_*), \mathbf{P}, \text{Adm, DAGSide, PetriSide,} \\ \text{SelectedRow, RegistryRow, LateStep, AgentSeam} \end{array} \right). \quad (3.62)$$

这里 c 只是在有限候选族中选择一个完整包，不是一个工程阶段。所有 seam 均绑定同一个 e_* 。特别地，

$$\begin{aligned} e_* &= \text{event}(\text{SelectedRow}) = \text{event}(\text{RegistryRow}), \\ \text{id}(e_*) &= (v, \text{rule, session, correlation, occurrence}), \end{aligned} \quad (3.63)$$

并且该 identity 在 replay、四个目标事件和 π -FMS seam 中逐坐标相等。加入独立的反馈证书后，定义完整认证模型

$$\text{Cert}(c, e_*) := \text{CoreCert}(c, e_*) \times_{\text{SelectedRow}} \text{FeedbackCert}(Q, U, K, H, \varepsilon). \quad (3.64)$$

这里的纤维积要求反馈证书所选正概率 row 的原生事件，正是核心证书中的 SelectedRow；因此组合两条主定理时不会悄悄换成另一条随机边。

3.12.2 Four-view coherence witness / 四视图相容见证

称 W 为 $\text{FourCoherence}(c, e_*)$ 的见证，当且仅当它同时包含：

1. 源与每个视图之间的有限路径 soundness 和从像出发的 reflection；
2. e_* 在四视图中的原生事件与 replay 端点；
3. success、external wait、deadlock、版本和资源坐标的保持；
4. DAG 的 SCC-internal/condensation-edge 二分与严格 SCC rank；
5. Petri 的 declaration delta、incidence、enabling、firing 和 provenance；
6. 强迟 π 原生导数、丰富 registry identity 与精确 Agent 端点；
7. strict admission 的目标恰为选中业务事件的源，且扩展方块相容；
8. 所有上述坐标共享式 (3.63)。

Main coherence theorem / 主相容性定理

对任意候选 c 与 occurrence e_* ,

$$\text{CoreCert}(c, e_*) \implies \exists W, \text{FourCoherence}(c, e_*; W).$$

因此同一业务发生在 DAG、Petri、强迟 π 、态射四个独立目标 LTS 中具有兼容的路径、反射、终态、版本、资源、准入与 replay 解释； π 行又与世界索引 Agent 域的精确端点一致。

3.12.3 Why the theorem closes / 定理为何闭合

证明按下列依赖顺序组合，且每一步只消费之前已经定义的对象。

1. 自由 SMC 普遍性质把静态 primitive 的解释延伸到全部协调图式。
2. 有限精确开放超图桥把静态接口放入粘合 DPO 环境；DPO 唯一性给出可执行端点。
3. 完整事件把端点构造压缩为 endpoint-free recipe；replay 唯一性固定同一 occurrence。
4. 商相容性使源 LTS 的路径与终态不依赖表示元。
5. 四个投影证书逐边给出 soundness、reflection 与 Lift，故扩展到任意有限路径。
6. 签名自然性运输旧语义，strict admission 单独证明新规则合法，二者共同闭合 epoch。
7. DAG 与 Petri sidecar 为同一 c_* 补充结构序和资源 provenance，而不替代全 LTS 证书。
8. 强迟 π 证书保留原生步与稳定身份；字面 seam 等式把该步连接至 Agent 域。

所以总定理不是“若若干图看起来相似，则相信它们一致”，而是从逐层可组合证书构造一个具体 W 。每个存在量都有来源，每个等式都有共同 occurrence 作为连接点。

3.12.4 Stabilization as a separate theorem / 独立的稳定化定理

四视图相容性描述“同一步在不同语义中是什么”；随机稳定化描述“调度长期会去哪里”。两者共用原生事件与 replay，却具有不同前提，因此必须保持为两条主结果。

Main stabilization theorem / 主稳定化定理

若 $\text{FeedbackCert}(Q, U, K, H, \varepsilon)$ 成立，则

$$\mu_{x_0}(\text{EventuallyAlways } U) = 1, \quad \mathbb{E}_{x_0}[T_U] \leq H/\varepsilon.$$

概率路径中的每条选中边均由源 LTS 原生事件标注，并可由其 recipe 精确重放。

Corollary: coherent stable coordination / 推论：相容且稳定的协调

一个完整认证模型同时具有四视图相容见证 W 与几乎必然稳定化结论。故在显式公平条件下，Cantilune 不仅能从任一视图追溯同一业务 occurrence，还能证明被调度执行最终进入并保持在稳定证据区；期望 epoch 数受 H/ε 控制。

3.13 Closed Theorem Ledger / 闭合定理账本

下表按数学依赖而非源码目录列出机械化理论的完整闭包。编号只用于本章引用，不承载项目阶段或交付计划含义。

Table / 表 3.5: 从变量与公理到主结论的定理闭包。

号	数学定理	主要前提	精确结论
T1	自由解释	A1, 目标 SMC 与 primitive compatibility	静态解释存在；指定原子分量后幺半自然同构唯一。

Continued on next page

Table / 表 3.5: 从变量与公理到主结论的定理闭包。(Continued)

号	数学定理	主要前提	精确结论
T2	DPO 两层唯一性	A2 , 同一 complement/right leg	范畴结果唯一到典范同构; 固定有限表示后认证端点相等。
T3	并行独立菱形	A2 , 显式 parallel independence	residual 均存在, 两种次序的终点典范同构; 不推出全局合流。
T4	签名扩展相容性	A1 、 A6	恒等与复合扩展在静态及四视图中自然; 纯 reindex 不等于 admission。
T5	商相容执行	A4	迁移、观察、终态和版本在状态商上良定义。
T6	事件重放唯一性	A2 、 A3	同一源与 endpoint-free recipe 产生唯一认证终点。
T7	投影路径双向性	A4 、 A5	源路径 soundly 映射; 从像出发的目标路径逐步反射并有 Lift。
T8	三个非名称视图闭包	A5 的 DAG、Petri、Morphism 证书	全 LTS 的路径、终态和版本在三个视图中保持并反射。
T9	Petri occurrence 等式	A7 、 A8	选中准入事件满足 declaration delta、incidence、enabling、firing 与 provenance。
T10	Open- π 范畴-操作桥	极化接口与结构同余	商 Hom 构成所需 SMC; plug、parallel、hide 在原生语义中闭合。
T11	有类型擦除	合法 typed late step	擦除得到真正 native untyped late step; 无反向类型恢复结论。
T12	最大相容域闭包	世界索引 ω CPO 与两分支各自前提	非分离域方程、hiding 与受限完全抽象; 分离伴随单独成立, 二者不合并。
T13	request-accept 精确证书	有限源演算与结构强迟 LTS	两条业务边强步 sound/reflect; 无 selected-candidate 因果前驱结论。
T14	丰富移动证书	A9 , 15 类事件族与 registry	每类与每登记位置有原生第一步; sound/reflect 并保持稳定 identity。
T15	硬前向不变量	有限高度 join 半格、 U 上闭、证据单调	进入 U 后所有有限执行都留在 U 。
T16	随机命中与 replay	A3 、 A10	几乎必然最终一直在 U , 期望 $\text{epoch} \leq H/\varepsilon$, 路径可原生重放。
T17	准入发生的执行分类	固定 occurrence 的完整执行语义	成功、外部等待、真死锁、稳定生产性无限轨迹构成互斥穷尽分类。
T18	候选索引四视图相容性	$\text{CoreCert}(c, c_*)$	存在 FourCoherence 见证; 同一 occurrence 连接四投影与 π -FMS seam。

表 3.5 的“依赖”是数学上的概念依赖, 并不声称最终机械化 声明在语法上逐字调用了前面全部定理。组合记录可以直接携带等价字段; 只要字段的 数学内容与此处一致, 闭环就成立。

3.14 Exact Scope, No-Go Boundaries, and Conclusion / 精确范围、不可行边界与结论

3.14.1 What is established / 已建立的结论

在 **A1–A9** 与完整候选证书下, 理论建立:

$$\text{CoreCert}(c, e_*) \implies \exists W, \text{FourCoherence}(c, e_*; W). \quad (3.65)$$

在额外的 **A10** 下，理论建立：

$$\text{Cert}(c, e_*) \implies \mu_{x_0}(\text{EventuallyAlways } U) = 1 \wedge \mathbb{E}_{x_0}[T_U] \leq H/\varepsilon. \quad (3.66)$$

两式合起来形成本章的数学闭环：

$$\begin{aligned} &\text{有限签名与静态组合} \implies \text{合法 DPO occurrence} \implies \text{唯一 replay} \\ &\implies \text{商上执行} \implies \text{四视图路径相容} \implies \text{名称与域端点相容}, \\ &\text{有限高度证据与公平 kernel} \implies \text{几乎必然永久稳定}. \end{aligned} \quad (3.67)$$

3.14.2 What is deliberately not claimed / 明确不作出的结论

1. DPO 并发定理不推出全局合流；只有显式 parallel-independent 的两步可交换。
2. 任意开放超图不存在保持全部 incidence 的 total strict-DAG 投影；总构造是 SCC condensation，产品级 strict-DAG 仍需证书。
3. 纯签名 reindex 不等于 strict admission；版本变化必须由准入事件证明。
4. 裸 π 迁移不能恢复审计元数据；丰富 identity 故意位于进程商之外。
5. typed erasure 只有正向操作保持，不提供任意 untyped derivative 的类型恢复。
6. request-accept 子语言与选中产品 occurrence 之间没有已证明的全局因果前驱关系。
7. separated domain branch 与 unseparated lower branch 不能拼成一个兼得所有性质的模型；后者的 effect bottom 合并 divergence 与 deadlock。
8. 概率结论需要稳定窗口、公平性、 $\varepsilon > 0$ 与 epoch 对齐，不是无条件活性。
9. 总定理是“完整证书蕴含相容见证”的条件定理，不声称任意候选或全部生产产品 已自动获得这些证书。

3.14.3 Proof status and interpretation / 证明状态与解释

与表 3.5 对应的核心义务在当前形式化清单中登记为 *kernel-proved*；项目状态仍为 *review-pending*，独立人工复核字段尚未完成 (Moonweave AI, 2026c,a)。形式化仓库还提供实质性参考见证以排除“所有类型都为空”的空洞闭包，但参考见证不是已部署生产包，也不能替代每个生产实例的完整证书。

Final interpretation / 最终解释

Cantilune 的数学本质不是四份图形化结果之间的事后比对，而是一个以完整 occurrence 为轴的证明携带语义：自由 SMC 固定可组合结构，DPO 固定合法变化，replay 固定同一端点，商 LTS 固定行为等价，四个投影证书固定路径与终态， π -FMS seam 固定名称传递的操作-指称意义，有限高度反馈固定长期稳定性。所有结论都沿同一依赖链返回 e_* ，因而变量有类型、公理有边界、定理有前提、理论有连接点、结论有精确范围。

Semantic architecture and certificates

语义架构与证书

4.1 Layers of the theory / 理论层次

Table / 表 4.1: The semantic layers and their roles in the current research branch.

Layer	Responsibility	What it does not establish by itself
Free SMC presentation	Legal typed sequential, parallel, and symmetric composition	A concrete scheduler or a projection's operational semantics
Typed open hypergraph / DPOI	Add, replace, reconnect, or delete a collaboration subgraph under stated boundary and gluing conditions	Global confluence, every proposed match, or every product rule
Projection certificates	Connect an identified source event to target transitions and replay evidence	A universal product instance without product-supplied certificates
Late- π layer	Sessions, request/accept, mobility, alpha conversion, and structural laws	A released channel API or unrestricted runtime protocol claim
Petri layer	Ordered interfaces and individual-token provenance	Boundedness, liveness, or numerical caps without net-level checks
Feedback / FMS boundary	Evidence closure, terminal classes, and the selected D1-A semantic route	A single unified source-paper FMS model or unrestricted full abstraction

4.2 Identified reconfiguration / 具备身份的重构

For a configuration G , a concrete rewrite event is represented schematically as

$$e = (\rho, m, \delta), \quad G \xrightarrow{e} G', \quad (4.1)$$

where ρ is the rule, m the chosen match, and δ retains derivation, representative, freshness, or allocation information. This is the minimum information needed to distinguish two applications of the same named rule and to define replay at the selected equality or isomorphism ([Moonweave AI, 2026g](#)).

4.3 Projection consistency target / 投影一致性目标

For each useful reading P_i , the intended condition is

$$G \xrightarrow{e} G' \implies P_i(G) \xrightarrow{\Phi_i(e)} P_i(G'), \quad i \in \{\text{DAG}, \text{Petri}, \pi, \text{morphism}\}. \quad (4.2)$$

The implication is a proof obligation, not a visualization convention. It must include the target event relation and appropriate preservation, reflection, replay, and terminal-classification clauses. The morphism reading is the identity view; the remaining readings require explicit certificates (Moonweave AI, 2026d,e).

Schematic event-certificate record

```

1 source event:      operation id, rule id, match, derivation, epoch
2 projection map:    target event(s), correspondence witness, replay identity
3 terminal view:     success | external wait | deadlock | productive-infinite
4 provenance:        source commit, manifest entry, build-evidence reference

```

4.4 Admission boundary / 准入边界

Dynamic graph change is intentionally constrained. The theory records structural, freshness, dangling, quiescence, resource, authorization, and replay conditions as applicable certificates. A future node or model may propose a new subgraph or channel, but the proposal is not equivalent to an accepted reconfiguration. This is the formal analogue of a visible control boundary, not a claim about an already deployed policy service.

Method, provenance, and reproducibility

方法、溯源与可复现性

5.1 Evidence model / 证据模型

The relevant reproducibility unit is not a dataset split or model checkpoint. It is a chain that binds a named theorem to an immutable source revision, a build/audit record, and a manifest entry. The current delivery record identifies source S , evidence E , and a later pointer/manifest revision P ; cleanup or documentation commits do not replace the proof identity of that chain (Moonweave AI, 2026a).

Table / 表 5.1: Evidence fields required to interpret a formal claim.

Field	Current record	Why it matters
Source branch	codex/theory-foundation	Fixes the reviewed repository state
Verified source S	59a1a688...e67d15dc	Identifies the Lean source tree behind the stated proof status
Evidence E	ed26cb74...3fb8dee5	Binds the build/audit record to the source claim
Manifest	formal/proof-obligations.json	Names declarations, status, provenance, and review fields
Build evidence	QA record bound to source S	Records the source-integrity and audit outcome
Review status	review-pending	Prevents a local or automated build from being reported as independent approval

5.2 Rebuild protocol / 重建协议

The formal project pins Lean and mathlib through its toolchain and lock files. Routine work must not silently update dependencies; the repository documents an ordinary kernel rebuild and a separate provenance-aware evidence gate (Moonweave AI, 2026b).

Repository verification commands

```
1 lake build
2 .\formal\scripts\ci.ps1
3 .\formal\scripts\ci.ps1 -RequireProved -VerifyTreeOnly
```

The commands are not an installation guide for an end-user product. The first rebuilds the kernel-checked project with the pinned toolchain; the evidence scripts verify source integrity, known dependencies, and the relationship between declared proof status and durable evidence.

5.3 Methodological exclusions / 方法上的排除项

No training corpus, benchmark dataset, model provider revision, inference hardware profile, or latency/cost experiment is part of the present evidence package. Such artifacts would become necessary only for a future product or runtime report. Reporting no empirical result is more accurate than repurposing template curves or synthetic measurements.

Risk / 风险

The status *proved* / *review-pending* is intentionally narrower than “validated product.” A successful local build is neither an independent review nor a substitute for FCP disposition, ADR acceptance, or per-product conformance evidence.

Mechanization and assurance

机械验证与保障

6.1 Formal project surface / 形式化项目范围

The repository’s `formal/` directory is a single Lean 4 proof project. Its lock file pins Lean and `mathlib`, while the proof manifest connects central declarations to source integrity, build evidence, and review metadata. The project gate rejects unknown kernel dependencies and does not permit project theorems to rely on `sorry`, `admit`, project-defined axioms, or unsafe declarations (Moonweave AI, 2026b).

Table / 表 6.1: Mechanized assurance surfaces relevant to the theory report.

Surface	Recorded artifact	Assurance purpose
Theory source	Cantilune Lean source tree	Defines the mechanized categorical, rewriting, π , feedback, and closure objects
Toolchain lock	Pinned Lean toolchain and resolved dependency lock	Makes the declared Lean and <code>mathlib</code> dependency state reconstructible
Proof manifest	Machine-readable central-obligation manifest	Names central obligations, Lean symbols, status, commits, and review fields
Source integrity	Source-integrity digest	Detects evidence/source mismatch before a proved claim is accepted
Build evidence	Durable QA build record	Preserves a durable audit trail rather than relying on a transient local cache

6.2 Current recorded scale / 当前记录规模

The branch README reports 565 maintained Lean source files, 1,624 audited declarations, and 18/18 central proof obligations recorded as *proved*; these values describe the repository evidence record, not a benchmark or a product reliability metric (Moonweave AI, 2026f,c).

565
Maintained source files
Lean source count in the branch record

1,624
Audited declarations
Recorded kernel-audit surface

18 / 18
Central obligations
Manifest status: proved

6.3 What the assurance gate means / 保障门的含义

1. **Kernel checking** establishes that the named Lean declarations elaborate under the pinned environment and the project gate's dependency restrictions.
2. **Provenance binding** establishes that a status refers to a specific source and evidence record rather than an unpinned working tree.
3. **Independent review** is not supplied by either of the first two steps. The manifest and delivery record retain explicit pending fields for reviewer and conflict-of-interest data.
4. **Product conformance** is a later obligation: a concrete package must supply its own rule, projection, admission, resource, authorization, fairness, and replay certificates.

No operational service claim

This section concerns the integrity of formal artifacts. It is not infrastructure evidence for availability, throughput, data protection, tool execution, or operational incident response, because no such runtime is in scope for the current branch.

Evidence and conformance status

证据与符合性状态

7.1 How to read the status / 如何解读状态

The central evidence is a proof-obligation manifest, not a benchmark suite. “Proved” means the named obligation is bound to the recorded source and build evidence; “review-pending” means the corresponding independent review and governance decision have not been recorded. The two labels must appear together wherever the current core theory is summarized ([Moonweave AI, 2026a,c](#)).

7.2 Recorded theory evidence / 已记录的理论证据

Table / 表 7.1: Current evidence statements and their deliberate boundaries.

Surface	Recorded evidence	Not established by that evidence
Generic core theory	18/18 central obligations recorded <i>proved</i> against the verified source and evidence chain	Independent QA-L4 review, FCP result, or ADR acceptance
P1a complete interface	A separate fixed-signature fourteen-event reference provides non-vacuity witnesses for supplied whole-LTS certificates	A certificate automatically generated for an arbitrary product rule family
P1c operation layer	Fifteen normative native event families and a sixty-operation registry have recorded strong late- π representatives	A public session API, provider integration, or runtime interoperability guarantee
D1-A boundary	Maximum-compatible D1-A closure records the selected lower-effect route and stated adequacy/full-abstraction scopes	A unified source-paper FMS model, unrestricted actual-Agent theorem, or reverse definability
Feedback layer	Finite-height closure, terminal classification, replay, and conditional hitting statements are part of the theory record	A production monitoring system

7.3 Consistency is an event-level claim / 一致性是事件级声明

For a recorded source change $G \xrightarrow{e} G'$, the useful claim is not merely that four dashboards can display related information. It is that every supplied projection certificate relates the same event identity to a target step and preserves the declared observation boundary. This is why the repository separates source events, target derivations, replay, terminal classes, and product certificates (Moonweave AI, 2026e).

$$\text{Replay}(G, e) = G' \quad \text{and} \quad \text{Replay}(P_i(G), \Phi_i(e)) = P_i(G') \quad (7.1)$$

Equation equation (7.1) is a target relation whose exact equality and certificate premises are defined by the formal artifacts; it is not an assertion that every future implementation can replay arbitrary external side effects.

7.4 No empirical scorecard / 不提供经验指标

The earlier runtime-template bars, latency curves, success rates, and safety-suite counts have been removed from this report. They had no evidentiary relation to Cantilune's current formal source tree. A future empirical report must name a released artifact, population, metrics, baselines, raw evidence, and uncertainty before it can make performance or safety claims.

Insight / 洞察

Formal status and empirical status answer different questions. A kernel-checked theorem can establish a stated mathematical consequence without establishing speed, usability, robustness, or real-world safety; conversely, a benchmark cannot replace a missing semantic proof.

Governance, review, and theory risk

治理、评审与理论风险

8.1 Governance boundary / 治理边界

The current formal evidence is deliberately governed as high-impact theory work. The specification, RFC, proof manifest, and delivery record distinguish automated proof checking from independent review and from a human FCP or ADR decision. A DRI can prepare evidence; that fact does not make the DRI an independent reviewer (Moonweave AI, 2026g,e,a).

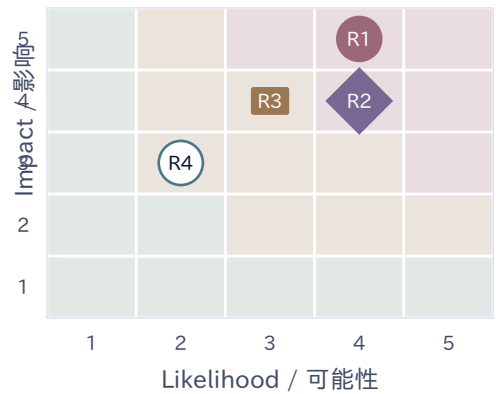


Figure / 图 8.1: Current pre-FCP theory risk register. R1: claim exceeds proved scope; R2: no independent review or FCP disposition; R3: product conformance remains uninstantiated; R4: source/evidence binding or toolchain verification becomes stale. Marker shape supplements color.

Source: Assessment derived from the current formal specification, proof manifest, and theory-closure delivery record (Moonweave AI, 2026g,c,a).

8.2 Current risk register / 当前风险登记

Table / 表 8.1: Risks that remain material even when the generic proof manifest reports *proved*.

Risk	Current control or evidence	Residual condition
R1: scope overclaim	This report labels all formal, product, review, and deployment boundaries explicitly	Readers may still infer runtime capability unless every public summary preserves the boundary
R2: non-independent review	Manifest and delivery record keep review evidence, reviewer identity, and conflict-of-interest fields explicit	Independent QA-L4 review and FCP disposition are pending
R3: product conformance gap	Generic certificate interfaces and reference witnesses are recorded; eight planned packages remain deferred	A concrete package can diverge unless its own rules and certificates are reviewed
R4: stale provenance	Verified source, evidence commit, manifest, and integrity gate are recorded	Later source, toolchain, or documentation changes require renewed binding and review

8.3 Review decision / 评审决策

Decision / 决策

Research-only status. The report accepts the generic theory record as *proved* / *review-pending*; it does not authorize deployment, production use, or a product safety claim. Any promotion requires independent review evidence, the applicable FCP and ADR outcome, and product-specific conformance certificates.

8.4 Responsible communication / 负责任的沟通

The theory explains how a future orchestration system could make authority, resources, sessions, and topology changes inspectable. It does not prove that a real deployment will prevent harmful actions, satisfy a policy regime, protect personal data, or meet a service-level objective. Those would require implementation, threat modeling, interface tests, operational controls, and domain-specific human review beyond the scope of this branch.

Limitations and non-claims

局限与非声明

9.1 Formal boundary / 形式边界

The formal specification and current closure record are intentionally precise about what they do not prove. The following limitations are part of the result, not footnotes to it.

Table / 表 9.1: Load-bearing limitations of the current Cantilune theory record.

Boundary	What is stated	What must not be inferred
Runtime absence	The formal core excludes runtime, LLM, I/O, tools, and network integration	That Cantilune already executes agents, tools, or workflows
Generic/product separation	Generic certificate interfaces and reference witnesses are present	That the eight planned production packages are instantiated or conformant
Projection scope	Certificates are parameterized by supplied target and product evidence; the morphism view is identity	That every arbitrary DAG, Petri, or π implementation automatically agrees with the source
DPOI / rewriting scope	Legal monic and boundary/gluing conditions are stated for the mechanized setting	Global confluence, critical-pair completeness, or validity of every proposed rewrite
D1-A FMS route	The maximum-compatible branch records specified lower-effect and contextual observation results	A single unified FMS model, constructor-sensitive strong-bisimulation full abstraction, or all-domain definability
Review status	Source/build evidence is bound for the generic core	Independent review, FCP passage, ADR acceptance, or production readiness

9.2 Operational limitations / 操作性局限

The formal object does not supply wall-clock bounds, cloud capacity, data governance, access control implementation, user experience, interoperability certification, or recovery procedures. It also does not turn a future model’s internal reasoning into a verified proof. A product report would need to specify the concrete rule schema, success predicate, policy boundary, resource model, external interfaces, and evidence collection before it could assess any of those concerns.

9.3 Interpretive limitations / 解释性局限

The phrase “one event across four views” has a strong meaning: a source event must retain a specified identity in its target counterparts. It does not mean that four visualizations or four logs are automatically consistent after the fact. Similarly, a reference execution shows that a certificate interface is nonempty; it does not prove that all applications of the interface are valid without their own supplied evidence (Moonweave AI, 2026d,e).

Risk / 风险

The most serious communication failure would be to report Cantilune as a released, safer, faster, or more capable agent runtime. None of those comparative or operational claims is supported by the current repository evidence.

Review gates and product path

评审门与产品路径

10.1 Theory promotion path / 理论推进路径

The next path is a review and conformance sequence, not a canary deployment. The current branch binds generic source and build evidence; subsequent gates test whether that record withstands independent scrutiny and whether a concrete package can instantiate the declared interfaces.

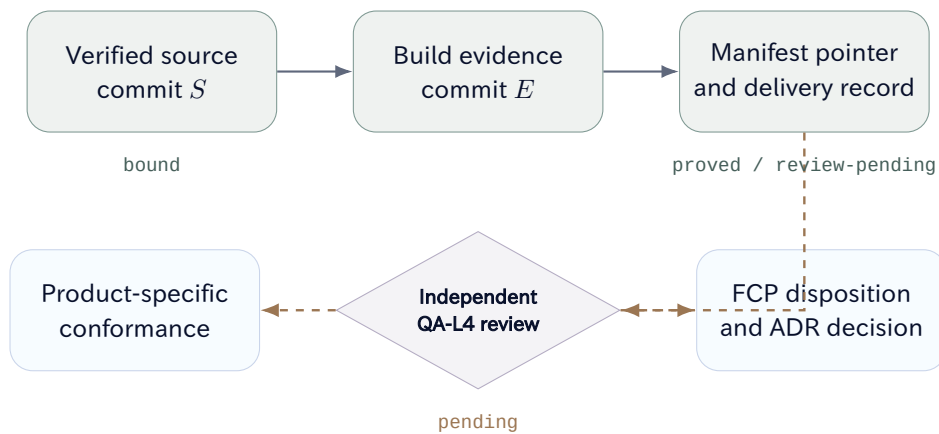


Figure / 图 10.1: Theory review path, not a product release plan. The current evidence binds a generic core to source and build evidence; independent review, FCP disposition, and concrete product conformance are subsequent and separate gates.

Source: Based on the theory-closure delivery record and formal-project build guidance (Moonweave AI, 2026a,b).

10.2 Gate register / 门控登记

Table / 表 10.1: Required gates after the current evidence-bound generic core.

Gate	Current status	Required exit evidence
Pinned rebuild and integrity audit	Recorded for the evidence-bound source	Reproducible build/ audit under the pinned toolchain and recorded provenance
Independent QA-L4 review	Pending	Reviewer independent of the DRI, review evidence, and recorded conflict-of-interest disposition
FCP decision	Pending	Formal review outcome against the stated core-theory scope and open boundaries
ADR disposition	Proposed / pending	Human decision that accepts, revises, or rejects the architecture boundary
Product conformance	Deferred	A package-specific rule schema plus projection, admission, resource, authorization, fairness, and replay certificates
Runtime validation	Not started	Concrete implementation, security model, external-interface tests, and empirical evidence appropriate to its claims

10.3 Change triggers / 变更触发条件

The evidence record should be re-evaluated when any of the following changes: the verified source commit, Lean or mathlib lock, proof manifest, evidence hash, central theorem scope, reviewer assignment, FCP/ADR status, or a product's declared operational rule family. A documentation-only change may preserve the existing proof identity, but it must not silently rewrite the status or scope communicated to readers.

No rollback protocol is claimed

There is no deployed Cantilune runtime to promote or roll back. The reversible operation here is a claim: if a source/evidence binding or review conclusion becomes invalid, the report must return to the narrower status rather than retaining an obsolete “proved” or “ready” summary.

Conclusion and next decisions

结论与后续决策

Cantilune proposes a single formal object for visible, evolvable orchestration: typed composition in $\mathcal{C}$, identified reconfiguration in $\mathcal{R}$, and certified readings into dependency, resource, communication, and compositional views. The current repository evidence binds a generic Core Theory to a verified source and build record; that is a meaningful formal milestone, but it is not a runtime or product launch (Moonweave AI, 2026f,a).

Decision / 决策

Current decision: retain pre-alpha research status. Publish the formal boundary and evidence chain as *proved* / *review-pending*; do not describe Cantilune as deployed, independently reviewed, FCP-approved, or product-conformant until those distinct records exist.

11.1 Prioritized next work / 优先后续工作

1. Obtain independent category/DPO/Petri, π /domain, and Lean/provenance reviews with explicit conflict-of-interest records.
2. Complete the applicable FCP and ADR decision path without widening the proof claims beyond their declared D1-A and projection boundaries.
3. Define one concrete product package and provide its complete per-rule conformance certificates, including admission, resource, authorization, fairness, and replay evidence.
4. Only after a runtime exists, prepare a separate implementation report for interfaces, security, usability, performance, reliability, and operational safety.

11.2 Reader checklist / 读者检查表

When reusing or extending this report, preserve the distinction between specification, mechanization, evidence binding, independent review, and product conformance. Update the source branch and commit identifiers before copying a theorem status, and remove any empirical or deployment language that is not supported by the target project's own artifacts.

Reproducibility record

可复现性记录

A.1 Evidence manifest / 证据清单

Table / 表 A.1: Reproduction record for the current theory claim.

Item	Recorded value
Repository	https://github.com/Moonweave-AI/cantilune/tree/codex/theory-foundation
Branch	codex/theory-foundation
Verified source S	59a1a6885ef6a2774b2731f487f83228e67d15dc
Evidence commit E	ed26cb74c4425b0d3025521f939695fd3fb8dee5
Pointer record P	0382b74074c546abe1bf3f37f3c03d7e4d2c3611
Normative prose	docs/spec/formal-semantics.md, RFC-0001, and RFC-0002
Proof manifest	formal/proof-obligations.json
Integrity record	formal/source-integrity.json
Toolchain and lock	formal/lean-toolchain and formal/lake-manifest.json
Build gate	lake build; formal/scripts/ci.ps1
Runtime / dataset	Not applicable: the current report has no released runtime, model, or empirical dataset
Review state	proved / review-pending

A.2 Verification interpretation / 验证含义

The ordinary build reconstructs the pinned Lean project. The provenance-aware gate checks source integrity and binds a proof status to the manifest and durable evidence. Neither command provides an independent review. A future reviewer should validate both the mechanical evidence and the scope statement attached to each central theorem (Moonweave AI, 2026b,c).

Status glossary and responsibilities

状态词汇与职责

B.1 Status glossary

Table / 表 B.1: Terms used consistently in this report.

Term	Meaning in this report
Specified	Defined in normative prose; it may still await proof, review, or implementation
Mechanized	Represented by a named Lean declaration in the formal project
Proved	Bound by the recorded manifest to the verified source and build evidence
Review-pending	No independent reviewer, FCP result, or ADR outcome has been recorded for the claim
Product conformance	A concrete package supplies and validates the certificates required by its own rules and policies
Runtime validation	Empirical and operational evidence for an implemented system; absent from this report

B.2 Responsibility model

Table / 表 B.2: Roles derived from the current formal documentation.

Role	Current responsibility
Owner / DRI	Joker-of-Gotham; owns the formal specification, evidence package, and stated scope
Formal contributors	Maintain Lean declarations, manifests, integrity records, and explicit theorem boundaries
Independent reviewers	Review category/DPO/Petri, π /domain, and Lean/provenance evidence independently of the DRI
FCP / ADR decision makers	Record the human governance decision; this cannot be inferred from a successful build
Future product owners	Instantiate package-specific rules, interfaces, policies, and conformance certificates

Publication checklist and change log

发布检查与变更记录

C.1 Publication checklist

Done	Check	Evidence
☐	Source branch, verified source, evidence commit, and pointer record match the cited artifacts	Document control and Appendix A
☐	Every theorem-status claim carries its review-pending boundary where applicable	Chapters 6 and 7
☐	No runtime, benchmark, safety-performance, or deployment claim appears without a matching artifact	Chapters 1 and 9
☐	Figures describe formal structure or review gates rather than fictional operational telemetry	Figures 2.1, 8.1, and 10.1
☐	Independent review, FCP, ADR, and product status stay pending until recorded	Chapters 8 and 10
☐	PDF build has no missing references, missing glyphs, or material overflow	Validation log

C.2 Change log

Table / 表 C.1: Report history.

Version	Date	Owner	Change
v0.1	2026-07-28	Joker-of-Gotham (DRI)	Re-aligned report to the codex/theory- foundation branch, including formal scope, proof provenance, governance boundary, and non-claims

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